How Sound Travels

Objectives:

1. To demonstrate that sound moves through different states of matter.

2. To observe that sound moves better through a solid than it does through a liquid or a gas.

Materials needed:

Demo: bell, tuning fork, cork, balloon, two spoons, musical triangle, slide whistle, two wood blocks, windup clock, 12" ruler, large jar filled with water, paper bag, metal can and lid, plastic container with lid, metal bar, music box motor.

Each group needs: meter stick, windup clock, self-sealing plastic bag filled with water, a block of wood (the same thickness as the plastic bag of water), two plastic cups with one end of a string through the hole in the bottom of one cup and the other end of the string through the hole in the bottom of the other cup. Knot the ends of the string to stop it from pulling out of the holes (home-made telephone), a third plastic cup with a string attached to the bottom of the cup, a metal rod.

Strategy:

Have the children identify different sounds made behind a screen (or desk) by using various materials: ringing a bell, hitting a tuning fork on a cork, blowing up a balloon and releasing air by stretching the neck of the balloon as the air escapes, tapping two spoons together, striking the musical triangle, blowing a slide whistle, hitting two wooden blocks together and ringing an alarm clock. Record the children's answers on the board. Show the materials used and check against the predictions made.

Hold a ruler over the edge of a desk and vibrate it asking such questions as: What is happening? What is your evidence? What is always happening to an object when it is making a sound?

Strike a tuning fork on a cork and quickly lower it into a glass of water. Ask the questions stated above. Does sound travel through other things?

Activities:

Demo: Does sound travel through paper? through plastic? through metal?

1. Put a ringing alarm clock in a paper bag.
2. Close the bag. Can you hear the clock ringing now?
3. Repeat steps 1 and 2 using a plastic container and a metal can.

Does sound travel through water?

1. Fill a glass jar nearly full of water.
2. Have one student cover one ear with her/his hand. Put the other ear against the glass jar.
3. Ask another student to hit two spoons together under water. Do you hear a sound?

Sound travels through different kinds of matter. Sound travels through gases, liquids and solids.

Group: What kind of matter does sound travel best in--air, water, wood or metal?

A. 1. Hold a plastic bag of water against one of your ears. Cover your other ear with your hand. Have someone hold a ticking clock against the bag of water. Listen.
2. Keep the clock in the same place. Remove the bag of water. Listen.
3. Place a block of wood between your ear and the clock. Listen. Do you hear the clock best through the air, the water or the wood?

B. 1. Place the clock 20 cm away from your ear and listen to the ticking.
2. Have your partner hold the clock at the 20-cm mark on the meter stick. Place your ear at the end of the meter stick and listen.
3. Have your partner hold a metal rod to your ear and place the clock against the rod 20 cm away from your ear. Listen. Does sound move differently through some solids than it does through others?

C. 1. Take the plastic cups (telephone) and hold your cup to your ear while your friend talks slowly and clearly into the other cup. Keep the string tight. How does it work? What is vibrating? How do the vibrations of your friend's voice reach your ear?
2. Can a third person talk and listen if another cup with a string is attached? Take the separate cup with the string and attach it to the first line. Keep all strings tight while one friend talks into one cup and the other two friends listen. Can you hear the message of the third party? How many lines could you attach? Does each addition weaken the vibrations?

Draw three squares on the board and in each square draw molecules spaced accordingly to illustrate the three states of matter: gas (molecules far apart), liquid (molecules closer together) and solid (molecules closest together). Have students identify the different states of matter for each square.

Finally, take the small music box motor, hold it in your hand and allow the music to play. Ask the students if they hear the music. Now place the music box motor on top of your desk or on a large surface area and listen to the melody. Is there any difference? Can the students hear the music now? Why?